AI Use Declaration Form

Course: Introduction to Artificial Intelligence
Lab Title: Lab 2: Predictive Analytics with Machine Learning
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GitHub Repository Link: https://github.com/JoshuaHanson69/25-26-sem1-cs212-a-assignment-1 JoshuaHanson69/blob/main/lab_2_predictive_analytics.ipynb
Date Submitted: 06/20/2026

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	**Yes**
Estimated percentage of the work influenced by AI	12%
Did you attach evidence of AI use where applicable?	**Yes**

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
ChatGPT	To help me understand what each lab task required, seek clarity on preprocessing steps, feature engineering, and interpret error messages encountered while coding as well as rephrasing interpretations.	can you explain what each TODO requires? What is the correct workflow for splitting data into training, validation, and test sets in this lab How do I choose evaluation metrics for regression and classification problems?	Conceptual understanding of preprocessing steps, feature engineering rationale, classification and clustering tasks.	I ran every cell myself and verified the outputs at each step before moving on. I compared AI explanations against my own understanding before writing any code

		how do i zoom in on the graph for better analysis		

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft	N/A	
Prompt history	Yes	AI_Use_Declaration_Form.docx
Screenshot of AI interaction	Yes	AI_Use_Declaration_Form.docx
Exported AI conversation	N/A	
AI-generated code snippet	Yes	AI_Use_Declaration_Form.docx
Revised version showing student input	N/A	

4. Student Declaration

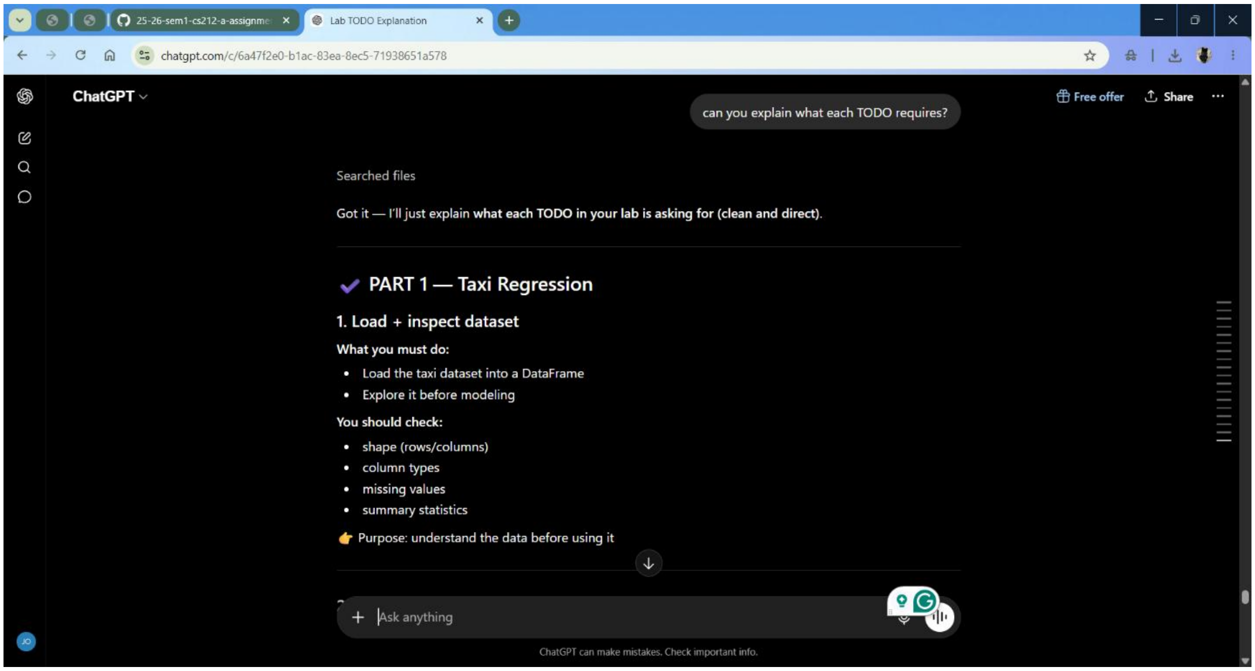
I declare that:

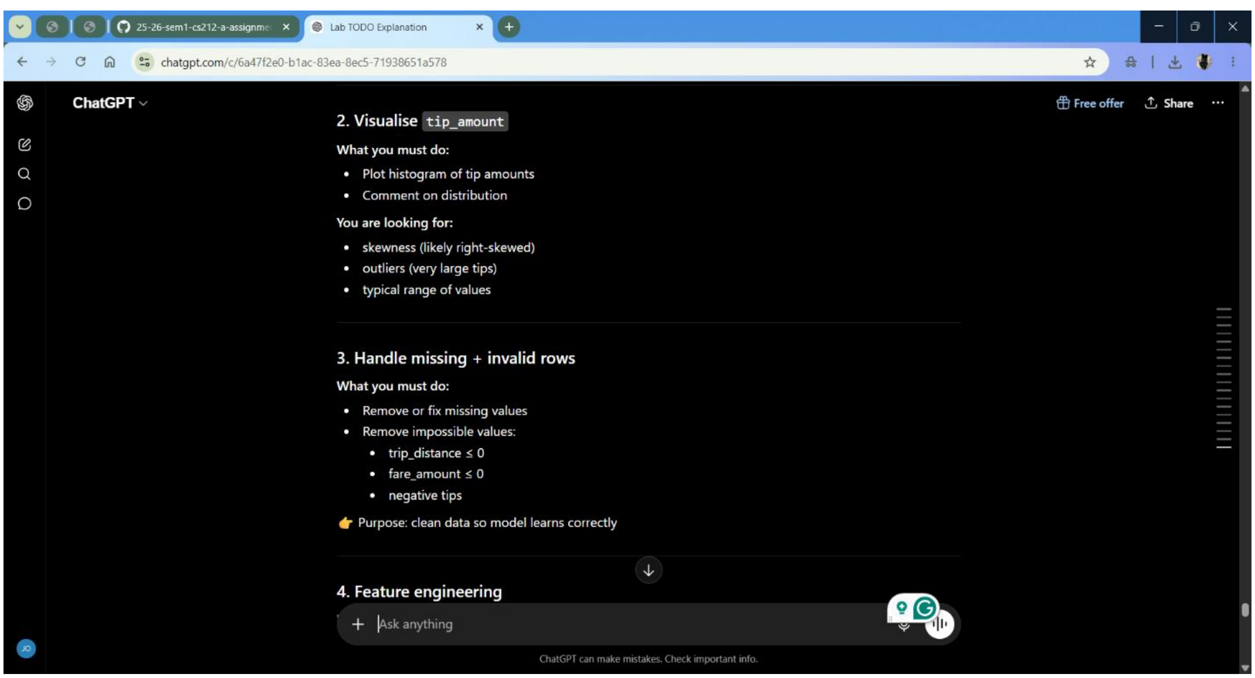
Declaration Statement	Response (Yes/No)
The submitted work is my own work.	YES
Any use of AI tools has been clearly declared in this form.	YES
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	YES
I reviewed, tested, edited, and improved any AI-generated content before submission.	YES
My AI usage does not exceed 25% of the entire work.	YES
I understand that undeclared or excessive AI use may be treated as academic misconduct.	YES

Student Signature: _____________J.E.H__________________________________

Date: ________________07/03/2026______________________________

SCREENSHOTS





4. Feature engineering

What you must do:

- Create at least 1 new useful feature

Examples of idea types:

- ratios (fare per mile)
- combined cost features

👉 Must explain why it helps prediction

5. Encode categorical variables

What you must do:

- Convert text/categorical columns into numbers

Choices:

- binary → 0/1
- nominal → one-hot encoding
- ordinal → ordered numbers (if applicable)

6. Scale features

What you must do:

- Apply `StandardScaler` to numeric features

9. Fit scaler only on training data

What you must do:

- Fit scaler using ONLY training set

👉 Purpose:

- avoid data leakage
- keep test set unseen

10. Train regression models

What you must do:

- Train at least 2 models (e.g. Linear Regression, Random Forest)

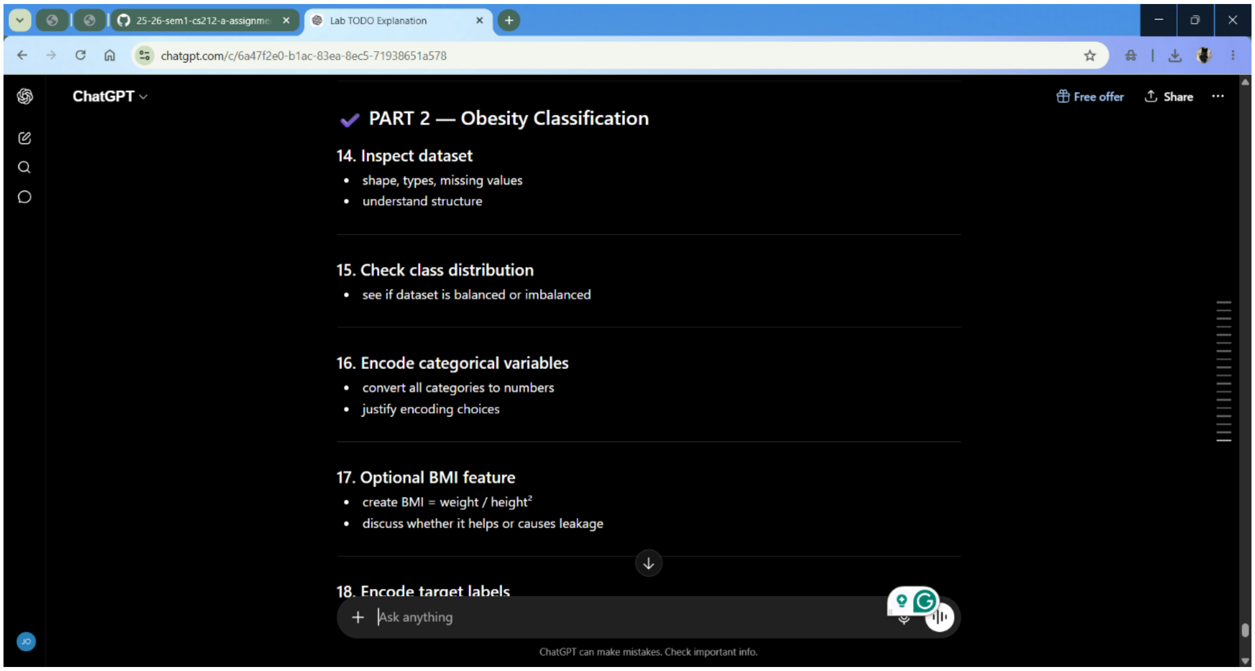
👉 Purpose:

- compare simple vs complex models

11. Make predictions

What you must do:

- Predict on train, validation, and test sets



PART 2 — Obesity Classification

14. Inspect dataset

- shape, types, missing values
- understand structure

15. Check class distribution

- see if dataset is balanced or imbalanced

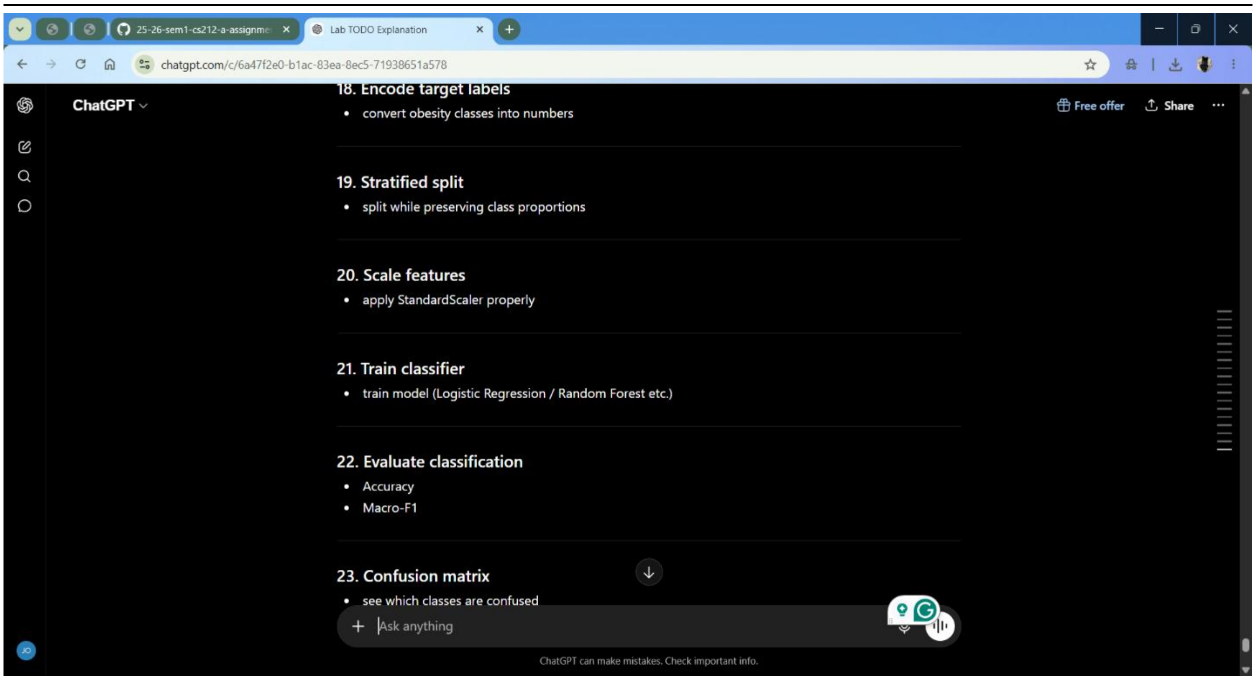
16. Encode categorical variables

- convert all categories to numbers
- justify encoding choices

17. Optional BMI feature

- create BMI = weight / height²
- discuss whether it helps or causes leakage

18. Encode target labels



18. Encode target labels

- convert obesity classes into numbers

19. Stratified split

- split while preserving class proportions

20. Scale features

- apply StandardScaler properly

21. Train classifier

- train model (Logistic Regression / Random Forest etc.)

22. Evaluate classification

- Accuracy
- Macro-F1

23. Confusion matrix

- see which classes are confused

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PART 3 — Clustering

24. Use only features (no target)

- remove obesity label
- clustering is unsupervised

25. Choose k

- use elbow method or silhouette score
- justify best number of clusters

26. Train KMeans

- fit model using chosen k

27. Visualise clusters

- reduce dimensions (or pick 2 features)
- plot clusters

Ask anything

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How do I handle missing values and invalid rows?

This TODO is asking you to make **your dataset clean and usable for modelling**.

Let's break it down into exactly what it means in your lab context.

1. Handle missing values (NaNs)

First question you should ask:

- Does my dataset contain missing values?

Check:

Python Run

```python
taxi.isna().sum()
```

What you can do (lab-appropriate options)

Option A: Drop missing rows (most comm for your lab)

Ask anything

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Python

```python
taxi = taxi.dropna()
```

When to use:

- only a small number of missing values exist
- dataset is large enough (taxi dataset is)

✔ Option B: Fill missing values (imputation)

Example:

Python

```python
taxi['fare_amount'] = taxi['fare_amount'].fillna(taxi['fare_amount'].median())
```

When to use:

- you don't want to lose data
- missing values are meaningful and frequent

✔ 2. Handle invalid rows (VERY important in this lab)

Now ask:

"Are there values that should NEVER exist in real life?"

For taxi data:

🚫 trip_distance ≤ 0

Python

```python
taxi = taxi[taxi['trip_distance'] > 0]
```

Why?

- A trip cannot be zero or negative distance

🚫 fare_amount ≤ 0

Python

```python
taxi = taxi[taxi['fare_amount'] > 0]
```

Why?

- fares should be positive (zero/negative = error or refund)

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🚫 tip_amount < 0

Python Run

```python
taxi = taxi[taxi['tip_amount'] >= 0]
```

Why?

- customers cannot give negative tips in normal cases

✔ Final combined solution (what your lab expects)

Python Run

```python
# 1. Remove missing values
taxi = taxi.dropna()

# 2. Remove invalid values
taxi = taxi[taxi['trip_distance'] > 0]
taxi = taxi[taxi['fare_amount'] > 0]
taxi = taxi[taxi['tip_amount'] >= 0]
```

✔ What this TODO is REALLY testing

Your lecturer is not just checking code — they want t[...]e if you understand:

Ask anything

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How do I choose evaluation metrics for regression and classification problems?

This is a **conceptual lab question**, and it appears in two specific parts of your notebook:

1. Regression task (Taxi dataset)

This relates to the TODO where you:

- evaluate models using **RMSE and R^2**
- compare performance on train, validation, and test sets

How to think about it:

For regression, you're predicting a **continuous number (tip_amount)**.

So you ask:

- How far are my predictions from the real values?
- Do I penalise large errors more?

So the correct metrics in your lab are:

Ask anything

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So the correct metrics in your lab are:

- **RMSE (Root Mean Squared Error)** → average prediction error size
- **R^2 score** → how much variance your model explains

Key idea:

You are measuring:

> "How close are my predicted numbers to the actual numbers?"

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2. Classification task (Obesity dataset)

This relates to the TODO where you:

- evaluate using **Accuracy and Macro-F1**
- interpret the confusion matrix

How to think about it:

For classification, you're predicting a **category (obesity level)**.

So you ask:

- Did I predict the correct class?
- Are some classes being ignored or poorly predicted?

So the correct metrics in your lab are:

Ask anything

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natural prompts you would have typed.

What is the correct workflow for splitting data into training, validation, and test sets in this lab

This question is pointing to the **data splitting TODO in your lab (both Taxi + Obesity sections)** where you are asked to create:

- Training set
- Validation set
- Test set

Correct workflow (what your lab expects)

Step 1 — Separate features and target

First you split:

- **X = features (inputs)**
- **y = target (what you are predicting)**

👉 For taxi:

- X = all columns except `tip_amount`

Ask anything

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Step 2 — Split into Train + Temp

You first split data into:

- Training set (largest portion)
- Temporary set (to be split again)

Typical:

- 60% train
- 40% temp

Step 3 — Split Temp into Validation + Test

Then you split the temporary set into:

- Validation set
- Test set

Typical:

- 20% validation
- 20% test

Final structure

Ask anything

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how do i zoom in on the graph for better analysis

To **zoom in on a graph for better analysis** in your lab, you have a few correct methods depending on what you want to see.

✔ Method 1: Clip the data (what you already did — BEST for this lab)

Python

```python
tip_zoomed = taxi['tip_amount'].clip(0, 40)
```

What this does:

- keeps only values between 0 and 40
- anything above 40 becomes 40
- anything below 0 becomes 0

Why this works:

You are not changing the plot — you are **zooming the data range itself**

✔ This is the most common method in data science

✔ Good for handling outliers

Ask anything

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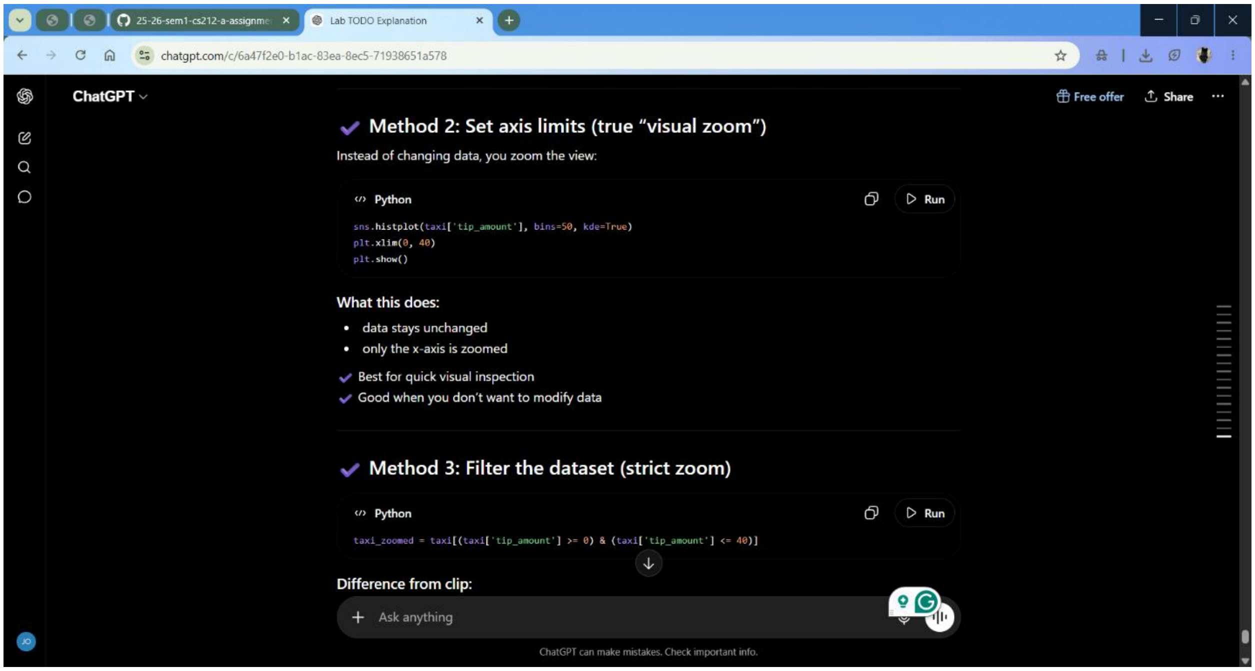

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Method 2: Set axis limits (true “visual zoom”)
Instead of changing data, you zoom the view:
Python
Run
sns.histplot(taxi['tip_amount'], bins=50, kde=True)
plt.xlim(0, 40)
plt.show()
What this does:
data stays unchanged
only the x-axis is zoomed
Best for quick visual inspection
Good when you don’t want to modify data
Method 3: Filter the dataset (strict zoom)
Python
Run
taxi_zoomed = taxi[(taxi['tip_amount'] >= 0) & (taxi['tip_amount'] <= 40)]
Difference from clip:
Ask anything
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